

Response to Reviewers

TimeSynth: a temporal fidelity benchmark for physiological signal forecasting

We thank the reviewers for a set of comments that were unusually specific and, in several places, correct about matters we had not checked ourselves. Acting on them led us to find three defects in our own code, one of which changes published numbers. We report those first, because they affect how the rest of this letter should be read.

Summary of the revision

What the revision adds	Scale
Validation on real recordings: PPG (BIDMC, PPG-DaLiA), ECG (MIT-BIH NSR, MIT-BIH AF), EEG (Sleep-EDF), two preprocessing tracks	420 trained models, 5 datasets, 3 modalities, 3 seeds
New baselines: TimesNet, TSMixer, CSDI (probabilistic), seasonal-naive floor	4 models added to the roster
Redesigned Markov paradigm with dwell-time control and pooled training	135 evaluations, D in {2, 5, 10} s
Tier 2 transient-rich generators (ECG with QRS, biphasic EEG spikes, PPG pulse)	3 corpora with exact event ground truth
Subject-level and signal-level statistics with non-overlapping test windows	14 paradigm tables, 3 seeds throughout
Adaptation arm: time-rescaling augmentation and few-shot fine-tuning	implemented

Corrections to the published results

1. Ground-truth leakage in the published MICN runs. Our data loader sets its own `label_len` to zero, so the target tensor contains only the forecast horizon. The training loop nevertheless builds the decoder input as `cat([batch_y[:, :args.label_len, :], zeros])`, and our MICN launch scripts passed `--label_len 50`. MICN was therefore given the first 50 samples, five seconds, of the ten-second target at training and at test time. We verified that those values are bit-identical to the ground truth. With the leak removed, MICN's mean absolute error on dual phase modulation rises from 0.0133 to 0.0712. All MICN numbers in the revision are leak-free, and the affected published values are corrected.

2. The transformer family never used time features. The embedding applied its temporal component only when the encoder input had exactly 125 steps. Every published run used a 50-sample history, so the encoder saw 25 steps and the branch was never taken. The Supplement is corrected to state that Transformer, Autoformer and TimesNet use value and positional embeddings only.

3. A figure used the wrong run. The frequency-shift analysis registered the single-phase-modulation FITS run under the drift-harmonic family. That curve is regenerated.

None of these affect the real-data experiments, which were trained from scratch under the revised protocol.

Response to Reviewers 1 and 2

R1.1 / R2.1: Omission of TimesNet and TSMixer

The claim that architecture determines fidelity is weakened by omitting TimesNet and TSMixer, which exist in the codebase; TimesNet directly tests the localized-processing hypothesis.

Response

We accept the substance and correct one point of fact. TimesNet was implemented and registered in the model table but never evaluated, which was an omission on our part. TSMixer was not in fact present in the codebase; there is no implementation of it in the released version. We have now implemented TSMixer following Chen et al., TMLR 2023, with time-mixing and feature-mixing blocks, residual connections and reversible instance normalisation, and we note in the Supplement that feature mixing is degenerate for univariate input. We agree that TimesNet is the sharper test of the locality hypothesis, since its two-dimensional convolution over inferred periods is an independent route to local processing, and it is now evaluated on every paradigm. We also added two baselines the reviewers did not ask for but which the comments imply: CSDI as a probabilistic baseline, and a seasonal-naive floor that repeats the last dominant period. The floor turned out to be one of the most informative additions to the paper, as described under R4.2 and R4.3.

Changes

Model/TSMixer.py added and registered; TimesNet, TSMixer, CSDI and the seasonal-naive floor evaluated with three seeds. TimesNet is expensive on our largest training sets, about 95 minutes per epoch, and a small number of its configurations did not complete within the cluster wall clock; these are listed as missing rather than imputed.

R1.2 / R2.2: PatchTST grouped with Transformers

The text attributes PatchTST's success to a localized bias, yet figures group it with Autoformer as a Transformer, conflating architecture family with inductive bias.

Response

The reviewers are right, and the inconsistency came from figures and text being maintained separately. We now group models by the operation that defines their inductive bias rather than by architectural lineage, and a single machine-readable file is the only source of group membership and colour for every figure and table, so text and figures cannot disagree again. PatchTST and TimesNet sit with the convolutional models under local receptive field; Autoformer joins DLinear, FITS and FreMLP under decomposition, since an explicit trend, seasonal or spectral split is what defines them.

Changes

configs/bias_groups.yaml added; all plotting and statistics scripts read membership and colours from it. Figures regenerated.

R1.3 / R2.3: Oversimplified physiological signals

Gaussian EEG spikes and ECG without R-peaks strip out the sharp, non-stationary transients that matter clinically, making the data overly homogeneous.

Response

We accept this and have restructured the generators into two declared tiers rather than defending the original ones as realistic. Tier 1 keeps the narrowband oscillators, because analytic phase and frequency ground truth requires narrowband components: a sharp QRS complex has no well-defined Hilbert phase, so adding one to the Tier 1 signals would corrupt the very ground truth the benchmark exists to provide. Tier 2 adds transient-rich signals and is evaluated with transient metrics instead of phase. The Tier 2 ECG uses the dynamical model of McSharry et al., 2003, with P, Q, R, S and T waves as Gaussian events on a limit cycle; wave amplitudes, widths and angular positions were fitted to MIT-BIH normal-sinus-rhythm beats with a box-constrained optimiser, with a median coefficient of determination of 0.97 across records. EEG transients are biphasic spikes, sharp waves and spike-and-slow-wave complexes with durations of 20 to 200 milliseconds, generated by a Poisson process. PPG uses a two-Gaussian pulse driven by the same beat-interval process as the ECG. Exact event times and amplitudes are stored with every signal, so peak metrics are exact rather than detector-dependent. We also now describe the signals as physiologically parameterised oscillators rather than as realistic physiology.

Changes

New generators, corpora and transient metrics (peak-timing error, peak-amplitude error, detection F1 within 50 milliseconds, inter-beat-interval error) are in place and unit tested, including a test that the stored event times match the waveform maxima to within one sample. [Numbers for the Tier 2 model sweep to be inserted once those runs complete.]

R1.4 / R2.4: Relative comparable-MAE window

The central-60 percent window is inflated by poor models such as Autoformer; within an absolute MAE window of 0.00 to 0.03, phase and frequency performance barely differ.

Response

The reviewers are correct, and this is the comment that changed our headline claim. We repeated the analysis at the level of individual sequences rather than model medians, binning every test window from every model by its error relative to signal amplitude, and also using the absolute window the reviewers propose. Within low-error bins the bias groups are separated by only a few degrees of phase, and the large separation appears only in the highest error bin. The reviewers' specific observation about Autoformer also holds: on drift-harmonic signals the gap between the best and worst bias group inside our original slab falls from 49.4 degrees to 1.1 degrees once Autoformer is excluded. We therefore no longer claim a dissociation at matched low error. The claim we can support is about the shape of the error distribution: models with local receptive fields produce near-perfect windows far more often and their failures stay bounded, whereas the other groups have a heavy tail of windows with phase errors near 60 to 90 degrees, which a mean pointwise error compresses and a phase metric does not.

Changes

Phase error in degrees by bias group, within bins of error relative to signal amplitude, all three signal families pooled:

relative MAE bin	Local receptive field	Global attention	Decomposition	Linear and MLP
0.00 to 0.05	1.05	1.58	0.39	0.16

0.05 to 0.10	3.18	2.64	1.38	3.16
0.10 to 0.20	5.84	3.30	6.83	6.47
above 0.20	12.39	65.49	63.65	54.81

The relative window moves to the Supplement as a sensitivity check, reported with and without Autoformer, and Results Q1 is rewritten in the narrower terms above.

R1.5 / R2.5: Zero-shot stress testing versus clinical practice

Deployed digital twins would use online updates, calibration, or augmentation, so penalizing static models for zero-shot extrapolation may not reflect clinical utility.

Response

We agree that a static zero-shot model is not what would be deployed, and we now present the shift experiments as a lower bound on deployed performance rather than as a prediction of it. To make that more than a wording change we implemented an adaptation arm: retraining with random time-rescaling augmentation that stretches the carrier frequency, and few-shot fine-tuning on five and twenty windows from the target shift bucket, measuring the fraction of the zero-shot gap each closes per bias group. This turns the objection into a result about which inductive biases adapt cheaply.

Changes

Augmentation and few-shot fine-tuning implemented in the runner and validated end to end. [Adaptation results to be inserted once those runs complete.] The shift sections are reworded as a lower bound regardless of that outcome.

R1.6 / R2.6: Frequency bounds in Appendix A2.3

Equation A7 gives f_0 in $[0.715, 0.861]$ Hz, but the text states $[0.71, 0.94]$ Hz; f_1 shows a similar mismatch.

Response

The reviewers are right and the error is in the text, not the code. Evaluating the generator's own expression gives f_0 in $[0.71485, 0.86145]$ Hz and f_1 in $[1.08135, 1.22795]$ Hz. The manuscript is corrected to these values.

Changes

Appendix A2.3 corrected. The bounds are now computed directly from the generator and written to a machine-readable constants file, so the text cannot drift from the code again.

Response to Reviewer 3

R3.1: Clinical relevance of single-signal phase

Clinicians care about timing delays and inter-signal phase relationships such as pulse transit time or cardiorespiratory coupling; single-signal phase is neither necessary nor sufficient for these.

Response

We accept that single-signal phase is not the clinically decisive quantity and have broadened the metric set rather than defending it. The real-data evaluation adds cross-correlation lag, band-power error, beat-timing error, inter-beat-interval error and beat-detection F1 within a 50 millisecond tolerance. We also added a clinical proxy: heart rate estimated from the forecast over the ten-second horizon, compared against heart rate from the observed future. On the BIDMC PPG recordings the best achievable error is 4.3 beats per minute, which is within the range used by consumer monitoring, whereas on ambulatory PPG and on ECG it is 12 to 44 beats per minute, which is not. Inter-signal phase genuinely requires synchronised multichannel recordings, which this benchmark does not yet include; we state that as a limitation and as the natural next step rather than claiming coverage we do not have.

Changes

Metrics added and reported for all real datasets; heart-rate proxy reported; the limitation on inter-signal phase is stated explicitly in the Discussion.

R3.2: Figure 8 implies the exact stochastic future must be predicted

Overlaying predicted and ground-truth futures suggests a good model reproduces one realization; the figure should show two futures sharing switching statistics.

Response

We agree, and the original figure did imply the wrong standard. The generator can now resume from the saved chain state and carrier phase at any forecast boundary and simulate independent continuations, so we store twenty alternative true futures for every test window. The replacement figure shows one history with two different realised futures that share the same switching statistics, the point forecasts regressing toward a blend of the two states, and samples from the probabilistic baseline. The caption states that the target is the distribution, not the realisation.

Changes

Figure 8 replaced; alternative futures stored with the corpus; the regression toward the mean is quantified as the fraction of forecast time whose local dominant frequency lies between the two state bands.

R3.3: Pooled or per-p training

A valid predictor must infer the switching probability from the observed history at test time; were models trained on pooled data or per p?

Response

They were trained per switching probability, which makes the reviewer's objection valid: the model could infer the switching rate from its training distribution instead of from the history. We confirmed this from the data layout and the checkpoint names, one checkpoint per probability. Training is now pooled over all dwell times, with the dwell time recorded in the metadata but never given to the model, and results are reported separately per dwell time at test time.

Changes

New generator writes a single pooled training and validation set with per-dwell test sets; all Markov results in the revision use pooled training.

R3.4: Switching probabilities are too high for oscillatory states

At 10 Hz with per-sample transitions, even $p = 0.5$ switches every other sample, destroying oscillatory cycles and producing noise-like signals.

Response

The reviewer is correct and this was a design error rather than a presentational one. The chain switched with probability p at every sample, so the mean dwell time was $1/(fs \cdot p)$ seconds, which for the published settings is far shorter than one cycle of the 1 Hz carrier. The paradigm has been regenerated with an explicit expected dwell time D in seconds and $p = 1/(fs \cdot D)$, with D in 2, 5 and 10 seconds, giving two to ten carrier cycles per state and roughly five, two and one expected switches inside the ten-second horizon.

Changes

Mean dwell time under the published parameterisation, at 10 Hz:

switching probability p	0.1	0.3	0.5	0.7	0.9
mean dwell time (s)	1.00	0.33	0.20	0.14	0.11
carrier cycles per state at 1 Hz	1.0	0.3	0.2	0.1	0.1

R3.5: Symmetric KL should compare transition probabilities directly

Gaussian KL is unmotivated; the transition matrices of the two fitted HMMs should be compared directly.

Response

We agree and have replaced the metric. We fit a two-state hidden Markov model to decoded features of the true futures and, separately, of the predicted futures, pooled across test windows for each dwell time, and report the symmetrised Kullback-Leibler divergence rate between the two chains, that is the stationary-weighted sum over states of the divergence between corresponding rows of the transition matrices. We also report the absolute error in estimated mean dwell time, which is easier to interpret. The Gaussian divergence is retained only in an archived script.

Changes

New metrics implemented and tested. They make the failure explicit: at an expected dwell of 10 seconds, against a probe-measured true dwell of 11.2 seconds, PatchTST predicts a dwell of 292.8 seconds, that is forecasts that essentially never change state, while having the lowest phase error of any model.

Response to Reviewer 4

R4.1: Overuse of digital twin

The work constructs no patient-specific, continuously updated replica and should be positioned as a forecasting benchmark.

Response

We accept this without reservation. The work is a forecasting benchmark with physiologically parameterised generators, not a digital twin, and the term is removed from the title, the abstract and the body except where we discuss what would be required to build one.

Changes

Title and framing changed throughout; the term is retained only in the Discussion, where it names future work.

R4.2 and R4.3: Methodological novelty and new scientific insight

The components are established techniques, MAE's limits are well known in signal processing, and local models doing well on periodic signals is unsurprising.

Response

We concede each of these points and have relocated the contribution instead of disputing them. We do not claim that mean absolute error is blind to phase, which is indeed textbook; we claim that forecasting evaluation still ranks models almost entirely by pointwise error, so the consequences go unmeasured, and we now measure them. The strongest new result comes from the real-data validation the reviewer asked for. Ranking models on the synthetic benchmark by mean absolute error carries almost no information about how they rank on real recordings, while ranking the same benchmark by phase fidelity transfers four times better:

Changes

ranking metric	mean Kendall tau, synthetic to real	negative transfers
mean absolute error	0.07	4 of 13 comparisons
phase error	0.28	1 of 13
frequency error	0.20	2 of 13

On real ECG morphology the transfer of the pointwise ranking is clearly negative, minus 0.58 and minus 0.47 against the two matched synthetic families, meaning the models that win on the synthetic benchmark are among the worst on the real signals. A synthetic benchmark is predictive of real performance only if it is scored on temporal fidelity. We report the moderate size of the positive correlation honestly: 0.28 is a useful signal, not a strong one.

Two further results speak directly to the charge that our findings are unsurprising. First, a seasonal-naive forecaster that simply repeats the last dominant period is simultaneously the worst model by mean absolute error and the best by phase error on real ECG, ranking last of sixteen on one metric and first on the other; on clean PPG it beats every one of the fifteen trained architectures outright, including on beat-detection F1 and heart-rate error. Second, on the redesigned switching paradigm the five best models by pointwise error occupy five of the six worst positions by switching fidelity. Neither result is predicted by the claim that local models win on periodic signals.

We also foreground the conditions where locality fails, as the reviewer invites. On Sleep-EDF electroencephalography the linear family leads and the naive floor is last, because there is no dominant period to repeat; under frequency shift the rank correlation between pointwise and fidelity rankings is approximately zero for every architecture family.

The contributions are rewritten as three specific claims: closed-form generators that make error decomposable into amplitude, frequency, phase and state; a quantified ranking disagreement reported as the headline result; and a map from inductive bias to the signal properties it preserves under each stressor. The local-models-win result is presented as a hypothesis the benchmark tests rather than as a conclusion.

R4.4: Signals are too simplified to be physiological

ECG lacks the QRS complex and PPG and EEG reduce to sinusoids; fitting parameter ranges does not make signals realistic.

Response

We agree, and our answer has two parts. The first is the Tier 1 and Tier 2 split described under R1.3, which adds QRS complexes and biphasic spikes and states the trade-off that forced the original simplification. The second is the real-data validation described under R4.5, which is the direct test of whether the simplification was made in the right places. The answer there is partly reassuring and partly not: fidelity-based rankings do transfer from the synthetic families to real recordings, with a mean rank correlation of 0.28, whereas pointwise rankings do not transfer at all. We report both.

Changes

Tier 2 generators and corpora added; the signals are described as physiologically parameterised oscillators; the transfer analysis is reported in full, including the comparisons where it fails.

R4.5: No validation on real data

Nothing shows the metrics or rankings transfer to real ECG, PPG, or EEG or to clinical events.

Response

This was the most consequential comment and the revision's largest addition. We now train and evaluate every model on five public datasets spanning three modalities: BIDMC and PPG-DaLiA for photoplethysmography, MIT-BIH normal sinus rhythm and MIT-BIH atrial fibrillation for electrocardiography, and Sleep-EDF for electroencephalography. Two preprocessing tracks are used, one band-limited to the dominant rhythm so that the window holds a comparable number of cycles to the synthetic benchmark, and one preserving morphology at a higher sampling rate. Splits are by subject, never by window, and all metrics are computed with the same implementations used on the synthetic data. Beyond the ranking-transfer result given under R4.2, the analysis covers clinically meaningful events: an atrial-fibrillation onset inside the observed history raises error by 53 percent and an activity change in ambulatory PPG by 56 percent, while sleep-stage transitions have no measurable effect, which is consistent with electroencephalography having no periodic structure for a transition to disrupt.

Changes

420 trained models across 5 datasets, 3 modalities, 2 tracks and 3 seeds, with per-subject metrics, the ranking-transfer analysis, the natural-event analysis and the heart-rate proxy reported in a new Results section and a supplementary table.

R4.6: Sample sizes and independence

The number of test signals, window overlap, and training runs should be reported; overlapping windows treated as independent would inflate significance.

Response

The reviewer identified a real problem. The original analysis enumerated test windows with a stride of one sample, so adjacent windows shared 149 of 150 samples and were treated as independent observations in paired tests. Every evaluation in the revision uses non-overlapping test windows, with a stride equal to the history plus the horizon, and the unit of analysis is the test signal for synthetic data and the subject for real data, with all windows averaged within a unit before testing. Every model is trained with three seeds. The primary test is a paired Wilcoxon signed-rank test over units against the linear baseline with Holm correction, reported with bootstrap confidence intervals and paired effect sizes. We re-ran the published comparisons under this protocol: of 99 model-versus-baseline comparisons that were significant at the window level, 80 remain significant at the signal level, and we identify the 19 that do not. We also increased the test fraction for the two primary real datasets, because a Wilcoxon test over three subjects cannot reach significance at any effect size.

Changes

A Methods table reports, for every paradigm, the number of units, windows per unit, window stride, overlap fraction, number of seeds and the test used. Overlap is zero throughout. A linear mixed model with random effects for unit and seed is reported as a sensitivity check.

R4.7: Markov switching is unsuitable for deterministic forecasters

Point forecasters cannot represent multiple stochastic futures, so their failure is expected; probabilistic methods are needed.

Response

We agree with the premise and have added CSDI, a conditional diffusion model, drawing fifty samples per window and computing switching statistics per sample. The result supports the reviewer's reasoning and also sharpens it. Deterministic models do not fail by producing a plausible but wrong realisation; they fail by refusing to change state at all. At an expected dwell of 10 seconds, against a probe-measured true dwell of 11.2 seconds, PatchTST predicts 292.8 seconds and NBeats 89.9 seconds, while the seasonal-naive floor predicts 15.8 seconds and has the lowest divergence of any model. Minimising expected pointwise error on a stochastic switching task is achieved by not committing to a state, and only a metric defined on the switching process exposes that. We report a negative result as well: our CSDI baseline does not recover the switching rate either, predicting a dwell of about 5 seconds regardless of whether the true dwell is 3 or 11 seconds. We therefore do not claim that probabilistic forecasting solves the problem, only that this diffusion baseline trained under our protocol does not.

Changes

CSDI added to the roster with per-sample switching statistics; Q4 retains the paradigm with the redesigned dwell-time parameterisation and reports both the deterministic and the probabilistic outcome.

Provenance of the numbers in this letter

Every quantity quoted above is produced by a script in the revision repository and written to a file that can be regenerated from the stored predictions.

Claim	Source
MICN leakage and the 0.0133 to 0.0712 change	analysis/p2_baselines/ACCEPTANCE_FAILURE.md, acceptance_seed2021.csv
Phase error by bias group within error bins; Autoformer sensitivity	analysis/p0_reanalysis/dissociation_groups.csv, dissociation_model_level_sensitivity.csv
Equation A7 bounds	analysis/constants.json
Synthetic-to-real rank transfer	analysis/p1_real/ranking_transfer.csv
Seasonal-naive ranks, heart-rate proxy, natural events	analysis/p1_real/supp_table_real.csv, hr_proxy.csv, natural_events.csv
Markov dwell times and divergence rates	analysis/p4_markov/markov_metrics.csv
Sample sizes, overlap, seeds and tests	analysis/p5_stats/sample_size_table.csv, paired_tests.csv
Tier 2 generator fit quality	Bio_Synthesize/Parametric_Fitting/tier2_ecg_fit.json